Labs 1 to 4 in Azure: your documents

AI Engineer (Data) course · Day 9 · in the browser, on your own

What you will build

Labs 1 to 4 built, by hand, a search that answers questions from FreshCart's 7 documents. Today you build the same thing from Azure's forms: Azure reads the files, cuts them into pieces, puts them in your own search index, and a Foundry agent answers questions from that index and names the file. Then you add one new file and see the agent answer from it, with no change to the agent.

> **7 files** → your own container (a folder in the class storage) → **knowledge source** (Foundry's link to that folder) and its index in Azure AI Search → **knowledge base** (the sources an agent may search) → **agent** → answers that name the file

Every part stands on its own unless it says what it needs.

Part (time)	Goal: what you have at the end, and what it needs
D1. Read the files with Document Intelligence (15 min)	You have seen Azure read a scanned PDF and a Word table, as Lab 1 did by hand. Browser only.
D2. Create a knowledge base and its source (15 min)	Your own index of the 7 files, which Foundry cut and indexed from your own container, as Lab 2 did. Browser only.
D3. Count and search your index (15 min)	You know how many pieces went into your index, and you have searched it with and without a filter, as in Lab 3. Needs your D2 index.
D4. Create an agent that answers from your files (15 min)	An agent that answers from your index, names the file, and refuses what the files do not cover, as in Lab 4. Needs your D2 index.
D5. Add a new file and re-index (10 min)	Your agent answers a question it could not answer before, after you add one new file to your container and your index takes it in. Needs D2 and D4.
Optional. The same in code (60 min)	D1 to D4 again as Python scripts, with a score on Labs 3 and 4's judged questions. Needs VS Code as in Lab 8.
Lab 9. Fabric pipeline (90 min)	A pipeline that copies only new tickets, checks and labels them, and runs in order. Its own handbook: Fabric_Lab9_pipeline.pdf.

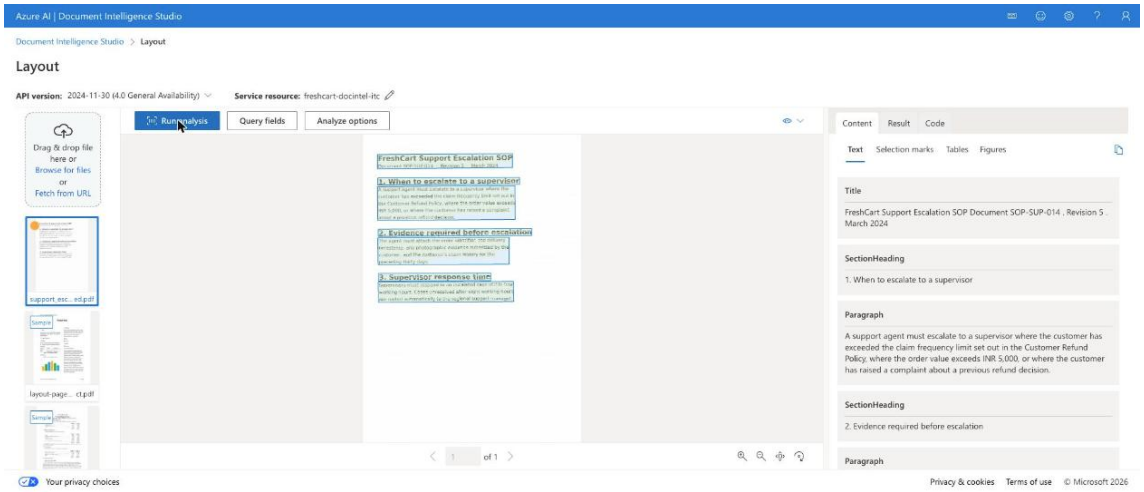
Before you start

- **Why this lab.** Labs 1 to 4 built a document search by hand on Databricks: read, cut, label, index, search, answer. Nothing there is wasted. Azure does the same steps for you behind a few forms, and you can only judge those forms because you built each step yourself. Watch for the step you recognise on each screen.
- **On your own, with your own names.** Everyone builds their own copy, signed in with their own class account. Your names come from your account: the account **itc_ai_1** gives **itc-ai-1**, so itc-ai-1-docs-source. Every name in this handbook uses itc-ai-1 as the example: **put your own in its place**. Write the underscores as dashes and keep everything in lower case; Azure refuses underscores and capital letters in these names. Only the agent's name has capitals: ItcAi1DocsAgent for itc_ai_1, ItcAi12DocsAgent for itc_ai_12.
- **Your search service.** The class has two search services, each holding 15 indexes, one per student. Accounts **itc_ai_1** to **itc_ai_15** use **freshcart-search-teams**; accounts **itc_ai_16** to **itc_ai_30** use **freshcart-search-teams2**. Make one knowledge source only. If it fails, delete it, then check in your search service, Indexes, that itc-ai-1-docs-source-index is gone before you make it again. If it is still there, tell your trainer: your service has no room for a second one.
- **Steps, then a table.** Each part has numbered steps to click. The table after them is what we found when we ran the same steps: read it and compare it with your screen. It has no steps of its own. The one exception is the results table at the end, which you fill in and hand in.
- **Where.** The class account you used in Lab 8. Foundry is **ai.azure.com**, project **itc-ai-trainer-0359**. The Azure portal is **portal.azure.com**, resource group **AIEngineerITC**. The 7 FreshCart files are in the **files** folder of labs1-4_azure; in D2 you upload them into your own container. D1 to D4 need no key.
- **This is your capstone path.** The capstone builds the same thing with your team's own files. The last section says what changes.

D1. Read the files with Document Intelligence (15 minutes)

Lab 1's loader returned 0 characters for the scanned SOP, and lost the Word table's column names. Here a reading service looks at each page as a picture and as text.

1. Find **support_escalation_sop_scanned.pdf** and **support_sla_matrix.docx** in the **files** folder of labs1-4_azure, or use your Lab 1 copy.
2. Open **contentunderstanding.ai.azure.com/documentintelligence/studio/layout** (Document Intelligence Studio, Layout) and sign in with the class account. When it asks for a resource, pick subscription **Subscription 1**, resource group **AIEngineerITC** and resource **freshcart-docintel-itc**, and confirm. The top of the page then says Service resource: freshcart-docintel-itc.
3. Drag **support_escalation_sop_scanned.pdf** into the box on the left, then **Run analysis**. Read the Content tab on the right: title, headings and paragraphs, read from a picture.
4. Do the same for **support_sla_matrix.docx**. Open the **Tables** tab: the columns have names.



The scanned SOP in Layout: every line found on the picture (left) and returned as text with headings (right).

What we checked (read only)	What we saw, and why it matters
The scanned PDF (Lab 1: 0 characters)	Read, with its title and 3 headings. (The optional code counts 825 characters.) Lab 1's loader returned nothing and raised no error: an empty file looks like a quiet success.
The Word table (Lab 1: column names lost)	Kept: Priority, Trigger, First response TAT and the rest, one row per priority. Lab 3's question R18 could not be scored without them.
The Lab 1 step this replaces	Load and parse. The service replaced your PDF and Word loaders, and added the reading of pictures (OCR) that Lab 1 did not have.

D2. Create a knowledge base and its knowledge source in Foundry (15 minutes)

In D2 you make two things, in this order. After them Foundry makes a third by itself.

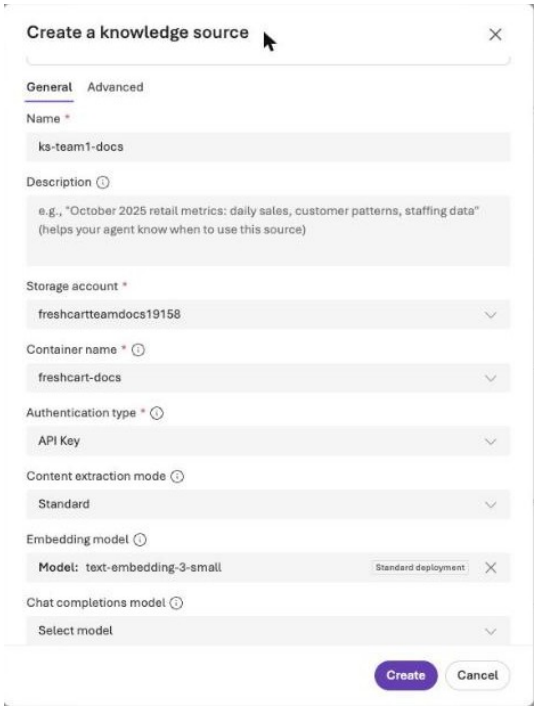
- **A knowledge base, itc-ai-1-knowledge-base.** A named group of one or more sources. It stores nothing itself. Your agent connects to it in D4.
- **A knowledge source inside it, itc-ai-1-docs-source.** It tells Foundry where your files are: your own container in the class storage.
- **Then Foundry builds the index, itc-ai-1-docs-source-index.** It reads the 7 files, cuts them into pieces, turns each piece into numbers that stand for its meaning (embeddings), and stores the pieces in the index on your search service, freshcart-search-teams or freshcart-search-teams2. This is the Lab 2 work, done for you.

  The search services are already made. freshcart-search-teams (itc_ai_1 to itc_ai_15) and **freshcart-search-teams2** (itc_ai_16 to itc_ai_30) are the class's Azure AI Search services, in resource group AIEngineerITC, and both are already connected to the Foundry project. The Knowledge and Agents lists show everyone's work: open and change only your own. You get only your own index, built through the form below, never a new search service. Each service holds at most 15 indexes, one per student, so make one source, and no spares.

  Your files go in your own container. The class shares one storage account, **freshcartteamdocs19158**. Inside it each student makes their own container, a folder for files, so your index reads only your files, and the

new file you add in D5 reaches no one else's index. You can do every step on your own, at any time.

1. In the Azure portal open the storage account **freshcartteamdocs19158**; in the menu on the left, **Data storage**, then **Containers**. Click **+ Container**, name it **itc-ai-1-docs**, **Create**. Open it, **Upload**, select all 7 files in the **files** folder of labs1-4_azure, **Upload**. Check that it lists 7 files. Do not add festive_season_addendum_2024.pdf yet: it belongs to D5.
2. In Foundry, project itc-ai-trainer-0359: **Build** (top bar), then **Knowledge** (left). If Foundry says **No access**, it opened another project: click **Select another project** and pick itc-ai-trainer-0359. The **Connection** box at the top must show your search service: **freshcart-search-teams** for itc_ai_1 to itc_ai_15, **freshcart-search-teams2** for itc_ai_16 to itc_ai_30. It starts at freshcart-search-teams, so accounts 16 to 30 open it and pick freshcart-search-teams2. Never click **Enable standard plan** in the grey bar: it opens a paid plan. If you open it by mistake, click **Cancel**.
3. **Create a knowledge base**. Name: **itc-ai-1-knowledge-base**. Leave the other boxes as they are.
4. Under Knowledge sources: **Add sources**, **Azure Blob Storage**. Fill in: name **itc-ai-1-docs-source**; storage account **freshcartteamdocs19158**; container **itc-ai-1-docs**, your own; authentication **API Key**; content extraction mode **Standard** (it starts at Minimal: change it); embedding model **text-embedding-3-small** (already filled). Leave the chat completions model empty. **Create**. Foundry names your index after the source, adding -index: **itc-ai-1-docs-source-index**.
5. Your source appears under Knowledge sources with the status **Creating**. On this page the status does not refresh by itself: wait 2 minutes, then click **Save knowledge base**.
6. **Check it worked.** Click **Knowledge** (left). Your row, **itc-ai-1-knowledge-base**, shows **itc-ai-1-docs-source** under Knowledge sources and **Active** under Status. If it says anything else after 3 minutes, check that your container lists 7 files, then delete the source (the ... on its row, Delete permanently) and add it again. The Connection box shows which search service it is on: if it is not yours, delete the source and tell your trainer.



The knowledge source form as the trainer filled it. Your name has your account name. The screenshot shows an older name, ks-team1-docs, and the shared container freshcart-docs; type yours, such as itc-ai-1-docs-source, and pick your own container, itc-ai-1-docs.

D2 is done when the source says Active and itc-ai-1-knowledge-base lists it. That shows the index was made, not what went into it: the first step of D3 counts that.

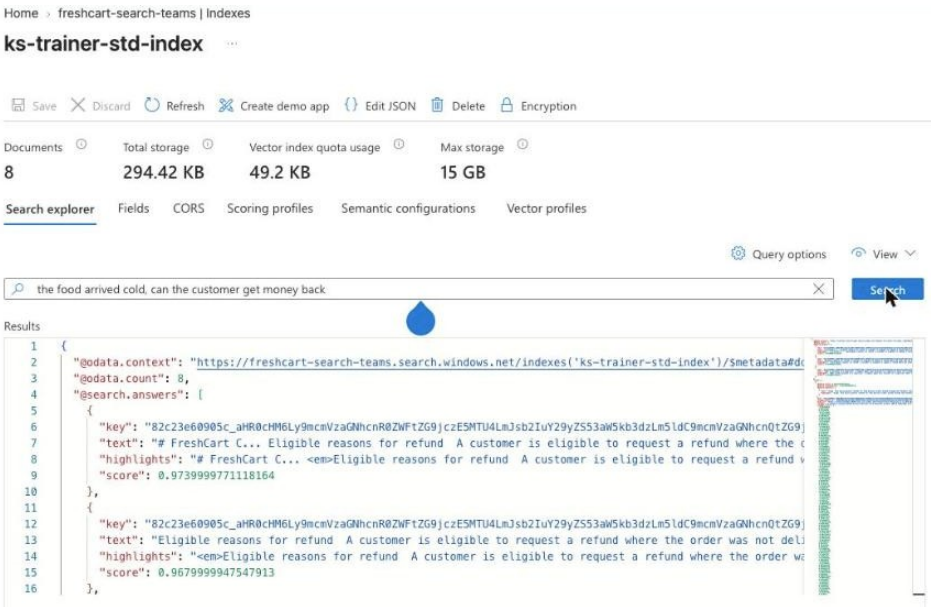
About this step (read only)	Why it matters
The Lab 2 steps these replace	Cutting (the form uses its own size, not your 500 characters), meaning numbers with an embedding model, and the index. Labels such as version and current-or-not are gone: the form does not know them. D3 shows what that costs.

Why API Key	The search service reads the storage with the storage account's key, which Foundry fetches for you. You never see or type it.

D3. Count and search your index in Azure AI Search (15 minutes)

1. In the Azure portal open your search service, **freshcart-search-teams** or **freshcart-search-teams2** (search for it in the top bar), then **Search management**, **Indexes**, and your index, **itc-ai-1-docs-source-index**. Check **Documents**: 8. This is the real test of D2. If you see fewer than 8, wait a minute and click **Refresh**; if it stays low, see the troubleshooting table at the end. (9 means festive_season_addendum_2024.pdf is already in your container: it belongs to D5, so see the troubleshooting table.)
2. In **Search explorer** type **the food arrived cold, can the customer get money back** and click **Search**. Read each result's **blob_url** (the file) and **@search.rerankerScore**, the reranker's score from 0 to 4.
3. Click **View**, **JSON view**. The explorer already sent your words twice: as keywords ("search") and as meaning ("vectorQueries"). That is Lab 3's stage 3, both merged, and the ready answers at the top come from the reranker, stage 4.
4. Now take the old policy out, as in Lab 3's stage 5, in the same search. Type the line below in Notepad and copy it. In the JSON query editor put the cursor just after the first **{**, press Enter, paste, then **Search**. The address is the old policy file's blob_url; put your own container name in place of itc-ai-1-docs:

```
"filter": "blob_url ne 'https://freshcartteamdocs19158.blob.core.windows.net/itc-ai-1-docs/refund_policy_v2_2022.pdf'",
```

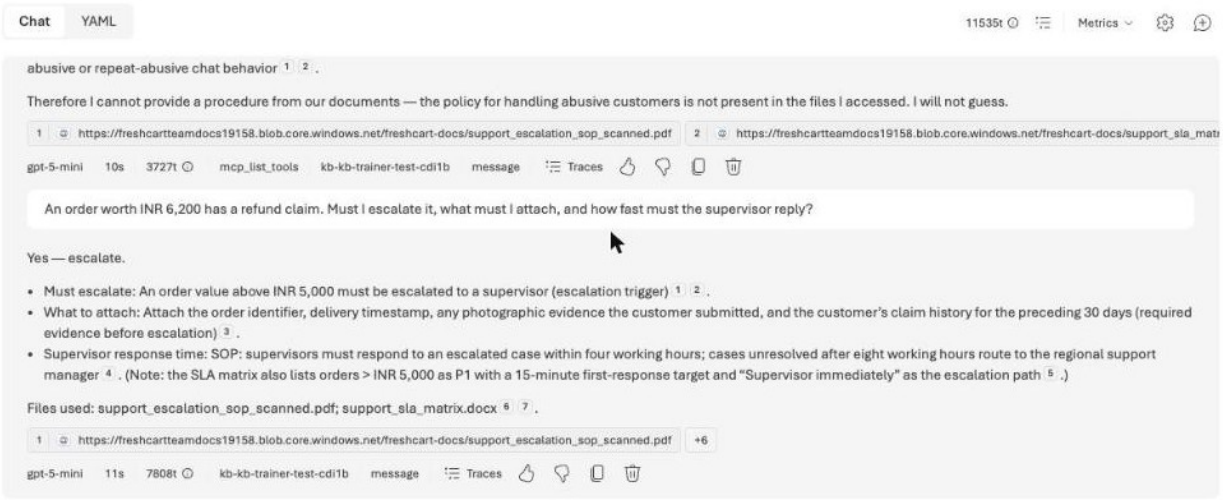


Search explorer on the trainer's index: 8 documents, the question, and the reranker's ready answers at the top.

What we checked (read only)	What we saw, and why it matters
Documents, with Standard (step 1)	8 pieces from 7 files. The scanned SOP is in, with its 3 headings kept.
Documents, with Minimal (trainer demo)	6 pieces from 7 files. The scanned SOP gave no text, and no error. Minimal reads only the text inside a file, so a scan is empty. It is Lab 1's silent failure again, now inside a form. Always count what went in.
The old policy	Without the filter the 2022 policy came second, almost level: 2.381 for the 2024 policy and 2.377 for the 2022 one. Similar is not correct. Read the scores under @search.rerankerScore. In Lab 2 a current-or-not label did this job; here the form made no labels, so you filter on the file's address instead.
With the filter	Gone. The top three became the 2024 policy, the support review (QBR) and the partner agreement, for the same question.
Four kinds of score	Keyword, meaning, merged and reranker scores are on different scales, as in Lab 2. Never set a cut-off on a score without knowing its scale.

D4. Create an agent that answers from your files (15 minutes)

1. In Foundry: **Build**, **Agents**, **New agent**, **Build an agent**. Name: **ItcAi1DocsAgent**. **Create agent and open playground**. Keep the model it starts with, **gpt-5-mini**.
2. Instructions: **You answer FreshCart support staff from FreshCart's internal documents only. Always name the file you used. If the documents do not answer the question, say so; never guess.**
3. Under Tools, open the **...** next to **Web search** and **Remove** it: answers must come from your documents. Under **Knowledge**: **Add**, **Connect to Foundry IQ**, then for connection pick your search service (freshcart-search-teams or freshcart-search-teams2), knowledge base **itc-ai-1-knowledge-base**, **Connect**. Then **Save**.
4. Ask: **An order worth INR 6,200 has a refund claim. Must I escalate it, what must I attach, and how fast must the supervisor reply?** Then ask: **A customer has been abusive on chat twice. What should the agent do, according to our procedures?**
5. Click a number in an answer to see the file it came from. Under the answer, **Traces** shows each step.



The trainer's test agent: the refusal (asked first, at the top), then the INR 6,200 answer and the files it used.

What we checked (read only)	What we saw, and why it matters
The INR 6,200 order	Escalate (over INR 5,000); attach the order ID, delivery time, photos and 30 days of claims; supervisor within 4 working hours. All from the scanned SOP, which Lab 4 could not read.
Two documents disagree	In our run the agent added that the SLA matrix makes the same order P1: first response in 15 minutes, supervisor immediately. In yours it may cite the SLA matrix and still give only the 4 hours, so the reader never learns the files disagree. Ask in the same chat: **Does the SLA matrix say anything different about this order?** Which rule wins is a business decision, not a search problem. (For discussion: who in FreshCart would decide it?)
The abusive customer	Refused: no FreshCart document covers it, and the agent said so instead of guessing. It still listed the two files it had searched.
The Lab 4 step this replaces	Answer, cite, refuse. Lab 4's code also checked that every cited piece came from the search. Here you check by clicking each number; the optional code does it for every answer.

D5. Add a new file and re-index (10 minutes)

Documents change after the index is built. FreshCart has just published a festive season addendum: new refund and escalation rules for the Diwali period. Your index and agent do not know it yet. This is the documents half of the syllabus's re-index pipeline: Lab 9 does the same for tickets.

1. Ask your agent ItcAi1DocsAgent first: **It is 2 November 2024. A customer claims INR 400 for a spilled curry. Who approves it and how fast is it credited?** Note what it says and which files it cites.
2. In the Azure portal open the storage account **freshcartteamdocs19158**; in the menu on the left, **Data storage**, **Containers**, and your container **itc-ai-1-docs**. **Upload festive_season_addendum_2024.pdf** from the

new_file folder of labs1-4_azure. Your container now lists 8 files.

3. Open your search service (freshcart-search-teams or freshcart-search-teams2), **Indexers**, **itc-ai-1-docs-source-indexer**. Look at its schedule: Foundry set it to run once a day. Click **Run**, then **Refresh** until the run shows Success. Read how many documents it processed: 1 is the new file. If it says 0, check that the file is in the container (step 2) and **Run** again. If it still says 0 and your index already counts 9, the daily run took the file first: go on to step 4.
4. Open **Indexes**, **itc-ai-1-docs-source-index**: count the documents again. Then ask the step 1 question again, and click the number in the answer.

What we checked (read only)	Why it matters
The run processed 1 document, not 8	The indexer remembers what it has read and takes only new or changed files: the same idea as Lab 9's watermark. Re-indexing a large library stays cheap.
The index went from 8 to 9 documents	Count after every re-index, as in D3 step 1. A file that fails to read shows up as a count that did not move.
Approved automatically, credited within 2 hours	Cited from festive_season_addendum_2024.pdf. The same agent, unchanged: only the index changed.
Trainer test, not a step: we deleted the file from storage and ran the indexer again. Still 9, and the agent still quoted it	Deleting a file does not take it out of the index: the indexer Foundry made does not look for deletions. Out-of-date rules keep answering until someone removes them, the same gap as the edited ticket in Lab 9. (For discussion: who in FreshCart should own that?)

Your results table (hand in)

	You	Our run	Labs 1 to 4 on Databricks
Your account (for example itc-ai-1)		trainer-demo	
D1: headings found in the scanned SOP		3	0 (no text)
D2: documents in your index (counted in D3)		8	21 pieces
D3: 2022 policy's place, without and with the filter		2nd, gone	2nd, gone
D4: the abusive-customer question refused? (yes / no)		yes	5 of 5 refused
D5: documents processed by the re-run, and the new count		1, then 9	re-run all by hand

What the class takes away

The steps of Labs 1 to 4 do not change in Azure: read, cut, index, search, answer, check. The forms do the work in minutes, but they hide choices you made by hand before: the piece size, the labels, which mode reads a scan. A form can drop a whole file without an error, and an old document still ranks second. Count what went in, filter what is out of date, and read the sources behind each answer.

For the capstone: your own files

Not yet tested with student accounts. The same four steps, with one change in D2. In the Azure portal open the storage account **freshcartteamdocs19158**, **Data storage**, **Containers**, **+ Container**, name it **itc-ai-1-capstone**, and upload your team's files into it. In D2 pick container **itc-ai-1-capstone** instead of itc-ai-1-docs, and name the source **itc-ai-1-capstone-source**. Each source makes its own index, and your search service holds 15, one per student, so first delete your own D2 source (itc-ai-1-docs-source for itc_ai_1): the ... on its row, Delete permanently. Keep files with personal data out of it: Lab 11 shows why.

Optional: the same in code (about 60 minutes)

For anyone who wants to see each step the forms hide, and score it as Labs 3 and 4 did. The scripts are in the **labs1-4_azure** folder and use a separate, shared index the trainer builds once (d2_index.py). Setup as in Lab 8: open the folder in VS Code, **Python: Create Environment** with requirements.txt, and a **.env** file with the four lines your trainer gives you: FOUNDRY_PROJECT_ENDPOINT, SEARCH_ENDPOINT, SEARCH_QUERY_KEY (it can only read) and DOCINTEL_ENDPOINT. You are signed in from Lab 8.

Run this	What it shows (our runs)
python d1_read.py	All 7 files read in one go; the scan gives 825 characters. Trainer demo: the free tier read 2 of the PowerPoint's 3 slides, with no warning.
d2_index.py (trainer)	24 pieces of 500 characters, each with Lab 2's labels: type, version, effective date, current or not.
python d3_search.py --rerank	Five ways to search, scored with nDCG@10 on Lab 3's 20 questions: 0.851, 0.876, 0.911, 0.920, 0.928 (Lab 3: 0.75 to 0.92); without --rerank, the first 3 only. The reranker made 3 questions worse. Run --rerank only if your trainer says so: each run uses 40 of the 1,000 free reranker searches a month on the shared service, so the class has room for about 20 runs in all.
python d4_answer.py	Lab 4's 25 questions: right facts 20 of 20 in all 4 runs. Refused, of 5: 5 of 5 in 2 of 4 runs. U01, "refund policy for grocery orders in Singapore", was answered in 2 runs, though the policy covers orders in India only (Lab 4 had the same trap). U04, about subscription fees, said the documents do not mention it, but not in the exact refusal sentence, so code counted it as an answer. Code checks every cited source against the search; **python d4_answer.py --fake** adds made-up sources, and 7 of 7 were flagged.

If something goes wrong

What you see	What to do
Studio asks for a resource you cannot find	Pick Subscription 1, then AIEngineerITC, then freshcart-docintel-itc. Signed in as another account? Sign out and in with the class account.
Knowledge source stays at Creating	On the page where you made it, it may stay at Creating until you save: save, then look at the Knowledge list. Still not Active there after 3 minutes: check the name is lower case with no spaces, delete the source and add it again.
Your index has fewer than 8 documents after a refresh	If it is 6, the mode was most likely Minimal: check the source's content extraction mode. If so, delete the source (the ... on its row, Delete permanently) and create it again with Standard. Any other count: check your container lists all 7 files, then open your indexer and click **Run** (D5 step 3 shows where).
Your index has 9 documents in D3	The D5 file is in your container too early. Delete it from your container, then delete your source (the ... on its row, Delete permanently) and make it again from D2 step 4.
A name is already taken	You may have typed the example, itc-ai-1, instead of your own account name. If your own is taken, someone has already built with your account: check the Knowledge list for it.
The agent answers from the web	Web search is still under Tools: remove it and Save.
Search explorer: the request is not valid	Paste the request from Notepad instead of typing it: the editor adds a closing brace while you type.

For the trainer

Point-and-click steps tested with the trainer's account on 27 October 2026; the optional code on 23 October 2026.

The class has two search services, freshcart-search-teams (itc_ai_1 to itc_ai_15) and freshcart-search-teams2 (itc_ai_16 to itc_ai_30), both Basic tier in West US 3, about USD 75 a month each while they exist. Each holds

15 indexes, one per student source, so the 30 students fill both. That leaves no room for your tests, the shared index the optional code reads (freshcart-docs-lab9) or Lab 11's freshcart-docs-lab11: build those on a third service, or keep one service under 15 students. freshcart-search-teams2 is not yet tested with a class account: check D2 step 2 with itc_ai_16 before class; in the capstone each student swaps their D2 source for a capstone one. Before class delete any test sources, knowledge bases and agents you made while trying the lab. The portal's own Import data wizard does not work here: it insists the search service read the storage with its own identity, which needs a role the class accounts cannot grant. Foundry's form uses the storage key instead. Each student makes their own container in D2 (itc-ai-1-docs for itc_ai_1) and uploads the 7 files from the pack, so students never affect each other and need no timing from you. The shared container freshcart-docs is for the optional code and your own demos only; name your own demo objects with trainer-demo, so they never take a student's name. After the course, delete the student containers and sources, and the search service freshcart-search-teams2.